Single particle GF

Free <u>fermi</u> gas $\quad H_0 = \sum_k \left(\frac{\hbar^2 k^2}{2m} - \mu\right) \hat{a}_k^+ \hat{a}_k = \sum_k \xi_k \hat{n}_k$

where $\hat{a}_k$ ($\hat{a}_k^+$) annihilates (creates) particle in state $|\vec{k}\rangle$

Coordinate representation $\rightarrow \hat{\psi}(x) = \frac{1}{\sqrt{V}} \sum_k \hat{a}_k e^{ikx}$

$$\hat{\psi}^+(x) = \frac{1}{\sqrt{V}} \sum_k \hat{a}_k^+ e^{-ikx}$$

annihilate (create) particle at point $\vec{x}$.

$$\hat{a}_k = \frac{1}{\sqrt{V}} \int dx\, \hat{\psi}(x) e^{-ikx}, \quad \hat{a}_k^+ = \frac{1}{\sqrt{V}} \int dx\, \hat{\psi}^+(x) e^{ikx}$$

number operator $\hat{a}_k^+ \hat{a}_k = \hat{n}_k$

$$\hat{N} = \sum_k \hat{a}_k^+ \hat{a}_k = \frac{1}{V} \int dx\, dx' \sum_k \hat{\psi}^+(x) \hat{\psi}(x') e^{ikx - ikx'}$$

$$= \int dx\, dx'\, \hat{\psi}^+(x) \hat{\psi}(x') \underbrace{\frac{1}{V} \sum_k e^{ik(x-x')}}_{\delta(x-x')}$$

$$= \int dx\, \hat{\psi}^+(x) \hat{\psi}(x)$$

$\Rightarrow \hat{\psi}^+(x) \hat{\psi}(x) = \hat{n}(x)$ is the density operator

$$H_0 = \int dx \left(\frac{\hbar^2}{2m} \frac{\partial \hat{\psi}^+}{\partial x} \frac{\partial \hat{\psi}}{\partial x} - \mu \hat{\psi}^+(x) \hat{\psi}(x) \right)$$

$$\hat{\psi}(x) \hat{\psi}^+(y) + \hat{\psi}^+(y) \hat{\psi}(x) = \frac{1}{V} \sum_{kk'} \left(\hat{a}_k \hat{a}_{k'}^+ e^{ikx - ik'y} \right.$$

$$\left. + \hat{a}_{k'}^+ \hat{a}_k e^{-ik'y + ikx} \right) = \frac{1}{V} \sum_{kk'} e^{ikx - ik'y} \underbrace{\left(\hat{a}_k \hat{a}_{k'}^+ + \hat{a}_{k'}^+ \hat{a}_k \right)}_{\delta_{kk'}}$$

$$= \frac{1}{V} \sum_k e^{ik(x-y)} = \delta(x-y)$$

Heisenberg representation : $\hat{A}_{SCHRODINGER} \longrightarrow \hat{A}_{HEIS} = e^{\frac{i\hat{H}t}{\hbar}} \hat{A}_{SCH} e^{-\frac{i\hat{H}t}{\hbar}}$

Thus $\hat{a}_k(t) = e^{+i\hat{H}_0 t/\hbar}\, \hat{a}_k\, e^{-i\hat{H}_0 t/\hbar}$

$$i\hbar \frac{\partial}{\partial t} \hat{a}_k(t) = -[\hat{H}_0, \hat{a}_k(t)]$$

Calculate $\underbrace{[\hat{H}_0, \hat{a}_k(t)]}_{\hat{H}_0 e^{\frac{i\hat{H}_0 t}{\hbar}} \hat{a}_k e^{-i\hat{H}_0 t/\hbar} - e^{\frac{i\hat{H}_0 t}{\hbar}} \hat{a}_k e^{-\frac{i\hat{H}_0 t}{\hbar}} \hat{H}_0 =} = \sum_p e^{i\hat{H}_0 t/\hbar} [\xi_p \hat{a}_p^+ \hat{a}_p, \hat{a}_k]\, e^{-i\hat{H}_0 t/\hbar}$

$$= e^{\frac{i\hat{H}_0 t}{\hbar}} [\hat{H}_0, \hat{a}_k]\, e^{-i\hat{H}_0 t/\hbar}$$

$$\sum_p \xi_p \left(a_p^+ \underbrace{a_p a_k}_{-a_k a_p} - a_k a_p^+ a_p\right) = \sum_p \xi_p \left(-\underbrace{a_p^+ a_k}_{\delta_{kp} - a_k a_p^+} a_p - a_k a_p^+ a_p\right) =$$

$$= \sum_p \xi_p \left(-\delta_{kp} a_p + a_k a_p^+ a_p - a_k a_p^+ a_p\right) = -\xi_k a_k$$

$$\Rightarrow [\hat{H}_0, \hat{a}_k(t)] = -\xi_k\, e^{i\hat{H}_0 t/\hbar} \hat{a}_k e^{-i\hat{H}_0 t/\hbar} = -\xi_k \hat{a}_k(t)$$

$$\Rightarrow i\hbar \frac{\partial}{\partial t} \hat{a}_k(t) = \xi_k \hat{a}_k(t)$$

$$\hat{a}_k(t) = e^{-i\xi_k t/\hbar} \hat{a}_k \; ; \; \hat{a}_k^+(t) = e^{i\xi_k t/\hbar} \hat{a}_k^+$$

$$\Rightarrow \hat{\psi}(x,t) = \frac{1}{\sqrt{V}} \sum_k \hat{a}_k(t) e^{ikx} = \frac{1}{\sqrt{V}} \sum_k \hat{a}_k e^{ikx - i\xi_k t/\hbar}$$

$$\hat{\psi}^+(x,t) = \frac{1}{\sqrt{V}} \sum_k \hat{a}_k^+ e^{-ikx + i\xi_k t/\hbar}$$

Time-ordered (causal, Feynman) GF

$$G^c(x_f, x_i; t_f, t_i) = -i \langle \hat{T}\, \hat{\psi}(x_f, t_f)\, \hat{\psi}^+(x_i, t_i) \rangle$$

Brackets $\langle \ldots \rangle$ denote average over the <u>ground state</u> of the system

$\hat{T}$ = time-ordering operator

$$\hat{T}\, \hat{\psi}(x_f, t_f)\, \hat{\psi}^+(x_i, t_i) = \begin{cases} \hat{\psi}(x_f, t_f)\, \hat{\psi}^+(x_i, t_i) & \text{if } t_f > t_i \\ -\hat{\psi}^+(x_i, t_i)\, \hat{\psi}(x_f, t_f) & \text{if } t_i > t_f \end{cases}$$

the minus sign is specific <u>for fermions</u>

<u>Compact notations</u> : let $x = (\vec{x}, t)$ denote both space and time

$$G^c(x, x') = -i \langle \hat{T}\, \hat{\psi}(x)\, \hat{\psi}^+(x') \rangle$$

In spatially homogeneous system G^c depends only on the difference $x - x'$, i.e. $\vec{x} - \vec{x}'$ and $t - t'$

$$G^c(x - x') = \int G(\vec{p}, \omega)\, e^{i\vec{p}(\vec{x} - \vec{x}') - i\omega(t - t')} \frac{d^3\vec{p}}{(2\pi)^3} \frac{d\omega}{2\pi}$$

Free fermi gas

$$G^c(\vec{x},t) = -i \langle \hat{T}\ \hat{\psi}(x,t)\ \hat{\psi}^+(0,0) \rangle =$$

$$= -\frac{i}{V} \sum_{kk'} \langle 0 | \hat{T}\ \hat{a}_k(t)\, e^{i\bar{k}\bar{x}}\ \hat{a}^+_{k'}(0)\, e^{-i\vec{k}'\vec{x}'} \overset{\vdots}{0} | 0 \rangle$$

For $\underline{t>0}$ $\quad \langle 0 | \hat{a}_k\, e^{i\bar{k}\bar{x} - i\xi_k t/\hbar}\ \hat{a}^+_{k'} | 0 \rangle =$

$$= e^{i\bar{k}\bar{x} - i\xi_k t}\ \langle 0 | \delta_{kk'} - \hat{a}^+_{k'} \hat{a}_k | 0 \rangle$$

$|0\rangle$ = ground state of Fermi gas: all states with $k \leq k_F$ are filled, states with $k > k_F$ are empty.

$$\Rightarrow \hat{a}^+_{k'} \hat{a}_k | 0 \rangle = \delta_{kk'}\ \hat{a}^+_k \hat{a}_k | 0 \rangle = \delta_{kk'}\ \hat{n}_k | 0 \rangle =$$

$$= \delta_{kk'}\ \theta(k_F - k)\ | 0 \rangle = \delta_{kk'}\ f_k\ | 0 \rangle$$

$$\Rightarrow \quad G^c(\vec{x}, t>0) = -\frac{i}{V} \sum_k e^{i\vec{k}\cdot\vec{x} - i\xi_k t} (1 - f_k)$$

$$f_k \equiv \theta(k_F - k) = \begin{cases} 1, & k_F > k \\ 0, & k_1 > k_F \end{cases} = \text{Fermi-Dirac distribution}$$

For $\underline{t<0}$ $\quad {}_{(-)}\langle 0|\, \hat{a}^{+}_{k'}\, \hat{a}_{k}\, |0\rangle\, e^{i\vec{k}\cdot\vec{x} - i\xi_k t} =$

$$= \delta_{kk'}\, f_k\, e^{i\bar{k}\bar{x} - i\xi_k t}(-1)$$

$$\Rightarrow G^c(\vec{x}, t<0) = \frac{-i}{V} \sum_k e^{i\vec{k}\cdot\vec{x} - i\xi_k t}(-f_k)$$

$$G^c(\vec{x}, t) = -\frac{i}{V} \sum_k e^{i\vec{k}\cdot\vec{x} - i\xi_k t}\Big((1-f_k)\,\theta(t) - f_k\,\theta(-t)\Big)$$

$$G^c(\vec{p}, \omega) = \int d\vec{x} \int dt\; G^c(\vec{x}, t)\, e^{-i\vec{p}\cdot\vec{x} + i\omega t} =$$

$$= -\frac{i}{V} \int_{-\infty}^{\infty} dt\; e^{i\omega t - i\xi_k t} \sum_k \underbrace{\int_{\vec{x}} e^{-i\vec{p}\cdot\vec{x} + i\vec{k}\cdot\vec{x}}}_{V\delta_{\vec{p}\vec{k}}} \Big((1-f_k)\,\theta(t) - f_k\,\theta(-t)\Big)$$

$$= -i\left\{(1-f_p) \int_0^{\infty} dt\; e^{i(\omega-\xi_p)t} - f_p \underbrace{\int_{-\infty}^{0} dt\; e^{i(\omega-\xi_p)t}}_{\substack{t=-t' \\ -\int_{\infty}^{0} dt'\, e^{-i(\omega-\xi_p)t'}}}\right\}$$

$$= -i\left\{(1-f_p) \int_0^{\infty} dt\; e^{i(\omega-\xi_p)t} - f_p \int_0^{\infty} dt\; e^{-i(\omega-\xi_p)t}\right\}$$

$$\int_0^\infty dt\, e^{ist} = \lim_{\delta\to+0} \int_0^\infty dt\, e^{ist-\delta t} =$$

$$= \lim_{\delta\to+0} \left.\frac{e^{ist-\delta t}}{is-\delta}\right|_0^\infty = \lim_{\delta\to+0} \frac{-1}{is-\delta} = i \lim_{\delta\to+0} \frac{1}{s+i\delta}$$

Thus, $G^c(\vec{p},\omega) = (-i)\left(\frac{1-f_p}{\omega-\xi_p+i\delta} - \frac{f_p}{-\omega+\xi_p+i\delta}\right) =$

$$= -i\left(\frac{1-f_p}{\omega-\xi_p+i\delta} + \frac{f_p}{\omega-\xi_p-i\delta}\right)$$

$$\begin{cases} f_p = \theta(k_F - p) \\ 1-f_p = \theta(p-k_F) \end{cases}$$

$$G^c(\vec{p},\omega) = -i\left(\frac{\theta(p-k_F)}{\omega-\xi_p+i\delta} + \frac{\theta(k_F-p)}{\omega-\xi_p-i\delta}\right) =$$

$$= -i\,\frac{1}{\omega-\xi_p+i\delta\,\mathrm{sgn}(p-k_F)}$$

$$\mathrm{sgn}(p-k_F) = \begin{cases} +1, & p>k_F \\ -1, & p<k_F \end{cases}$$

$$\xi_p = \frac{\vec{p}^2}{2m} - \mu = \frac{\vec{p}^2}{2m} - E_F, \quad k_F = p_F = \sqrt{2mE_F}$$

Observables

Density at $\vec{x}$: $n(\vec{x}) = \langle \hat{n}(\vec{x}) \rangle = \langle \hat{\varphi}^+(x)\, \hat{\varphi}(x) \rangle =$

$$= -i \lim_{t_i \to t_f + 0} G^c(\vec{x}, \vec{x}; t_f, t_i)$$

$$= -i \lim_{t' \to t+0} \lim_{\vec{x}' \to \vec{x}} G^c(\vec{x}, \vec{x}'; t, t')$$

Note that $G^c(\vec{x}, \vec{x}'; t, t') = \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{d\omega}{2\pi} G(\vec{p}, \omega) e^{i\vec{p}(\vec{x}-\vec{x}')} e^{-i\omega(t-t')}$

$$\Rightarrow n(\vec{x}) = -i \lim_{\tau \to +0} \int \frac{d^3\vec{p}}{(2\pi)^3} \int \frac{d\omega}{2\pi} G(\vec{p}, \omega)\, e^{i\omega\tau}$$

$t' - t = \tau = +0$